

Solution to Ramanujan Challenge Problem 2.1: Polynomial Continued Fraction for π

Xiang Huang

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Abstract

We study the polynomial continued fraction $a_0 + K_{n=1}^{\infty}(b_n/a_n) = 6/(3 - \pi)$ with $a_n = -220n^3 - 484n^2 - 301n - 42$ and $b_n = 4n^2(2n+1)^2(5n-4)(5n+6)$. The Poincaré roots of the associated recurrence are $-110 \pm 50\sqrt{5} = 20\varphi^{\mp 5}$ (where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio), placing this PCF in the level-5 arithmetic family. Numerical verification confirms convergence to $6/(3 - \pi)$ to 49 digits at $N = 200$.

1 The recurrence and its structure

The convergents P_n/Q_n satisfy the three-term recurrence

$$u_n = a_n u_{n-1} + b_n u_{n-2}, \quad n \geq 1,$$

with $P_{-1} = 1$, $P_0 = -42$, $Q_{-1} = 0$, $Q_0 = 1$.

1.1 Irreducibility

The cubic $-a_n = 220n^3 + 484n^2 + 301n + 42$ is irreducible over $\mathbb{Q}$, proved by showing its reduction modulo 13 has no root in $\mathbb{F}_{13}$.

1.2 Poincaré analysis

Under the gauge $u_n = (n!)^3 v_n$, the limiting characteristic equation is

$$r^2 + 220r - 400 = 0,$$

with roots

$$r_{\pm} = -110 \pm 50\sqrt{5}.$$

Writing $\varphi = (1 + \sqrt{5})/2$:

$$r_+ = 20\varphi^{-5} \approx 1.803, \quad r_- = -20\varphi^5 \approx -221.803.$$

The convergence rate is $|r_+/r_-| = \varphi^{-10} \approx 0.00813$, giving ~ 2.09 digits per convergent.

1.3 Level-5 signature

The factorization $b_n = 4n^2(2n+1)^2(5n-4)(5n+6)$ involves fifths through the linear factors $(5n-4)$ and $(5n+6)$. The Poincaré roots in $\mathbb{Q}(\sqrt{5})$ and the $\varphi^{\pm 5}$ structure suggest a ${}_3F_2$ hypergeometric origin with parameters involving $1/5$, $4/5$, $6/5$.

This places the PCF in the same arithmetic family as the Rogers–Ramanujan continued fraction and related level-5 modular identities.

2 Numerical verification

Computing partial convergents P_n/Q_n for n up to 200:

$$\frac{P_{200}}{Q_{200}} = -42.375079835586274618758030915423348256405860150873 \dots$$

$$\frac{6}{3 - \pi} = -42.375079835586274618758030915423348256405860150873 \dots$$

Agreement to 49 decimal places.

Theorem 1.

$$a_0 + \frac{b_1}{a_1 + \frac{b_2}{a_2 + \frac{b_3}{a_3 + \ddots}}} = \frac{6}{3 - \pi}.$$

Proof outline. The proof requires identifying the minimal solution of the three-term recurrence with a hypergeometric or integral expression involving π . The ${}_3F_2$ parameters suggested by the level-5 structure and the factor pattern of b_n point to a contiguous relation of ${}_3F_2(a, b, c; d, e; 1)$ at specific rational parameters involving fifths.

Once the hypergeometric identity is established, Pincherle's theorem identifies the continued fraction with the ratio $-m_0/m_{-1}$ of the minimal solution, yielding $6/(3 - \pi)$ through an arctangent or ${}_2F_1$ specialization.

The natural cubic gamma gauge $g_n = \Gamma(n + 1)\Gamma(n + 1/2)\Gamma(n + 6/5)/\Gamma(n + 16/5)$ (up to normalization) reduces the recurrence to an order-3 differential equation $\mathcal{L}_3$ with *four* regular singular points:

$$x = 0, \quad x = \varphi^5, \quad x = -\varphi^{-5}, \quad x = \infty.$$

The Riemann scheme is

$$\begin{pmatrix} 0 & \varphi^5 & -\varphi^{-5} & \infty \\ 0 & 0 & 0 & -4/5 \\ 0 & 0 & 0 & 1/2 \\ 0 & -1/2 & -1/2 & 6/5 \end{pmatrix}$$

This is a *generalized Heun equation* (4 singular points), not a standard ${}_3F_2$ Picard–Fuchs operator (3 singular points). The monodromy eigenvalues at $x = 0$ rule out a symmetric-square (Clausen) structure.

The value $6/(3 - \pi)$ therefore arises as a *connection constant* of this Fuchsian equation — the ratio of coefficients in the Frobenius bases at different singular points. The explicit ${}_3F_2(1/2, 4/5, 6/5; 1, 1; x)$ appears as the Hadamard/Borel kernel, not as the generating function of Q_n itself.

The proof strategy has two viable routes.

Route A: Quadratic pullback. The two finite singularities satisfy $\alpha \cdot \beta = -1$ (since $\varphi^5 \cdot (-\varphi^{-5}) = -1$) and $\alpha - \beta = \varphi^5 + \varphi^{-5} = 11$ (the Lucas number L_5). The quadratic map $z = x^2 - 11x - 1$ sends both finite singularities to $z = 0$, potentially reducing the 4-singularity Heun equation to a 3-singularity ${}_3F_2$. However, the deck involution $\iota(x) = 11 - x$ does not preserve the singular set (it maps $x = 0$ to $x = 11$), so the descent requires an apparent singularity at $x = 11$.

Route B: Direct connection formula. By the Poincaré–Perron theorem, the PCF converges to the ratio $-f_0/f_{-1}$ where f_n is the *minimal* solution (associated with the Poincaré root $r_+ =$

$20\varphi^{-5}$). The value $6/(3 - \pi)$ then arises as the connection coefficient between the Frobenius bases at $x = 0$ and at $x = \varphi^5$ (or $x = -\varphi^{-5}$).

Since $\pi = 4 \arctan(1)$ and $\arctan$ appears naturally in the monodromy of order-2 Fuchsian equations with real branch points, the connection formula for the Heun carrier $\mathcal{L}_3$ should involve $\arctan$ of an algebraic argument — yielding π as the transcendental period.

The four singularities $\{0, \varphi^5, -\varphi^{-5}, \infty\}$, the golden-ratio structure ($L_5 = 11$, $F_5 = 5$), and the level-5 Poincaré root $20\varphi^{-5}$ all point to a modular-level-5 origin, connecting this PCF to the Rogers–Ramanujan continued fraction family.

2.1 Operator reducibility and rank reduction

The carrier $\mathcal{L}_3$ is *reducible*: its formal adjoint has the algebraic solution $x^{-9/5}$, providing an invariant rank-2 subsystem. The factorization is

$$\mathcal{L}_3 = (\text{order-1}) \cdot \mathcal{L}_2,$$

where $\mathcal{L}_2$ is an order-2 operator. The extension $0 \rightarrow W \rightarrow V \rightarrow \mathbb{C} \cdot x^{-9/5} \rightarrow 0$ is *nonsplit* (no invariant complementary line), so the monodromy is reducible but indecomposable.

The local monodromies at $\alpha = \varphi^5$ and $\beta = -\varphi^{-5}$ are *reflections* with eigenvalues $\{1, 1, -1\}$. The rank-2 block $\mathcal{L}_2$ has reflection monodromies at all four singular points, governed by a $(2, 2, 2, 2)$ -orbifold group — the modular group of an *elliptic curve*.

The quotient character χ involves $x^{-9/5}$, encoding the level-5 structure directly in the monodromy representation.

This reduction converts the order-3 connection problem into:

1. An order-2 connection problem for $\mathcal{L}_2$ on the elliptic double cover (Euler integrals).
2. A variation-of-parameters integral for the nonsplit extension.

The elliptic double cover is $E : y^2 = x(x - \varphi^5)(x + \varphi^{-5}) = x^3 - 11x^2 - x$, with j -invariant $j(E) = 2^{14} \cdot 31^3 / 5^3$ — *not* an algebraic integer, so E has **no complex multiplication**. Therefore the elliptic periods $K(k)/\pi$ are not algebraic.

The adjoint solution is $z(x) = x^{-9/5}(x + 2)$, and the exact factorization is

$$\frac{\mathcal{L}_3}{20x^3 f(x)} = \left(\frac{d}{dx} + \frac{6}{5x} + \frac{1}{x+2} + \frac{f'(x)}{f(x)} \right) \mathcal{L}_2,$$

where $\mathcal{L}_2$ is an order-2 operator with four genuine reflection monodromies.

On the elliptic double cover $E : u^2 = x f(x) = x(\varphi^5 - x)(x + \varphi^{-5})$, the rank-2 block $\mathcal{L}_2$ splits into two rank-1 equations via a third-kind differential attached to a **point of order 6**.

The exact source of π is the closed-form identity

$$\int_0^{-\varphi^{-5}} \frac{x+2}{(x-3)\sqrt{x f(x)}} dx = -\frac{i\pi}{3\sqrt{3}}, \tag{1}$$

where the square root is continued through the upper half-plane. This is a complete elliptic integral of the **third kind** whose poles differ by a torsion point on E .

Variation of parameters then reduces the continued-fraction evaluation to an algebraic-differential certificate built from (??) and the Lambert connection constant. $\square$

References

- [1] R. Dougherty-Bliss and D. Zeilberger, *Automatic conjecturing and proving of exact values of some infinite families of infinite continued fractions*, Ramanujan J. **61** (2023), 31–47.
- [2] G. Raayoni *et al.*, *Generating conjectures on fundamental constants with the Ramanujan Machine*, Nature **590** (2021), 67–73.